

CSS Beginner Handbook (Part 1)

Margin, Padding, Percentages & Gradients

Chapter 1: Margin

Margin is the transparent space outside an element's border. It separates elements without changing their internal size.

Why use it? It improves readability, creates whitespace, and prevents elements from touching each other.

Properties:

- margin
- margin-top
- margin-right
- margin-bottom
- margin-left

Accepted values:

- px – fixed spacing
- % – relative to parent width
- auto – browser calculates margins (commonly for centering)
- negative values – pull elements closer/overlap

Example:

```
div{
  width:300px;
  margin:auto;
  margin-top:20px;
}
```

Chapter 2: Padding

Padding is the space between content and the border. Unlike margin, the background color extends into the padding area.

Properties:

- padding
- padding-top
- padding-right
- padding-bottom
- padding-left

Accepted values:

- px
- %
- em
- rem

Padding increases the visible box size unless `box-sizing:border-box` is used.

```
button{
  padding:12px 24px;
  background:steelblue;
  color:white;
}
```

Chapter 3: Percentages

Percentage values are relative, not fixed.

Width percentages are based on the parent's width.

Height percentages usually require the parent to have an explicit height.

Use percentages to create responsive layouts.

```
.parent{width:800px;}
.child{
  width:50%;
}
```

Chapter 4: CSS Gradients

Gradients are generated images used mainly as backgrounds.

Types:

1. Linear Gradient – straight line.
2. Radial Gradient – spreads from a center.
3. Conic Gradient – rotates around a center.
4. Repeating variants of all major gradient types.

Useful values:

- to right
- to bottom
- 45deg
- circle
- ellipse
- color stops (red 20%, blue 80%)

```
/* Linear */
background:linear-gradient(to right,#4facfe,#00f2fe);

/* Radial */
background:radial-gradient(circle,yellow,orange,red);

/* Conic */
background:conic-gradient(red,yellow,green,blue);
```

Next Parts: Display, Position, Flexbox, Grid, Units, Backgrounds, Borders, Typography, Transitions, Animations, Responsive Design, etc.